

- Abstract
- 1. Introduction
- 2. Multi-Property Synthesis: Multi-attribute LTLf Synthesis
 - 2.1 Core Ideas
 -
 - 2.2 Symbolization Advantage
 - 2. 3 Performance Comparison
- 3. Proc3D: Procedural 3D Generation and Parametric Editing
 - 3.1 Procedural Compact Graph (PCG)
 - 3.2-3 Generation and Editing Process
- 4. SpaceHMchat: Human-Machine Collaborative Health Management for Spacecraft Power Systems Performance
 -
 - 4.1 System Principle
 - 4.2 Process Flow
 - 4.3 Experimental Verification
- 5. Comprehensive Comparison
- 6. Future Directions
- References

Abstract

The combination of formal methods and generative models offers new possibilities for complex system engineering. This paper reviews three cutting-edge works: Multi-Property Synthesis proposes a multi-property LTLf synthesis method, which avoids enumeration by computing the relationship between states and achievable goal sets through a single fixed-point calculation; Proc3D introduces the Procedural Compact Graph (PCG) representation to achieve the generation and real-time editing of editable 3D models from text; SpaceHMchat builds a human-machine collaboration framework and applies it to the full-loop health management of aerospace power systems. By comparing and analyzing the core ideas, technical routes, and experimental results of each method, this paper explores the application potential of formal methods and generative models in safety-critical fields such as aerospace. With the rapid development of artificial intelligence technology, how to combine the rigor of formal methods with the flexibility of generative models has become a hot topic of concern in both academia and industry. In safety-critical fields such as aerospace and robotics, systems need to simultaneously meet temporal logic constraints and possess efficient generation capabilities. This review selects three papers that explore this intersection from different perspectives: Multi-Property Synthesis focuses on efficient algorithms for multi-objective LTLf synthesis; Proc3D utilizes large language models to achieve the programmatic generation of editable 3D models; SpaceHMchat introduces human-machine collaboration into the health management of aerospace power systems. This review aims to summarize the core contributions of these works and look forward to future development directions.

2. Multi-Property Synthesis: Multi-attribute LTL_f Synthesis

Multi-Property Synthesis addresses the issue that traditional LTL_f synthesis can only handle a single objective or requires the enumeration of subsets by proposing a method for computing all achievable objective combinations in a single fixed-point calculation.

2.1 Core idea

Transform multiple LTL_f properties into their respective $DFAs$ and construct the product automaton. Then, by introducing Boolean target variables $\{k_1, \dots, k_n\}$ to represent subsets of properties, define the initial winning relation:

$$w_0(Z, K) = \bigwedge_{i=1}^n (k_i \rightarrow f_i(Z_0))$$

Using the standard fixed-point calculation with symbols (the same as for a single attribute), we obtain $w(Z, K)$, which compactly characterizes the relationship between each state and the set of achievable goals.

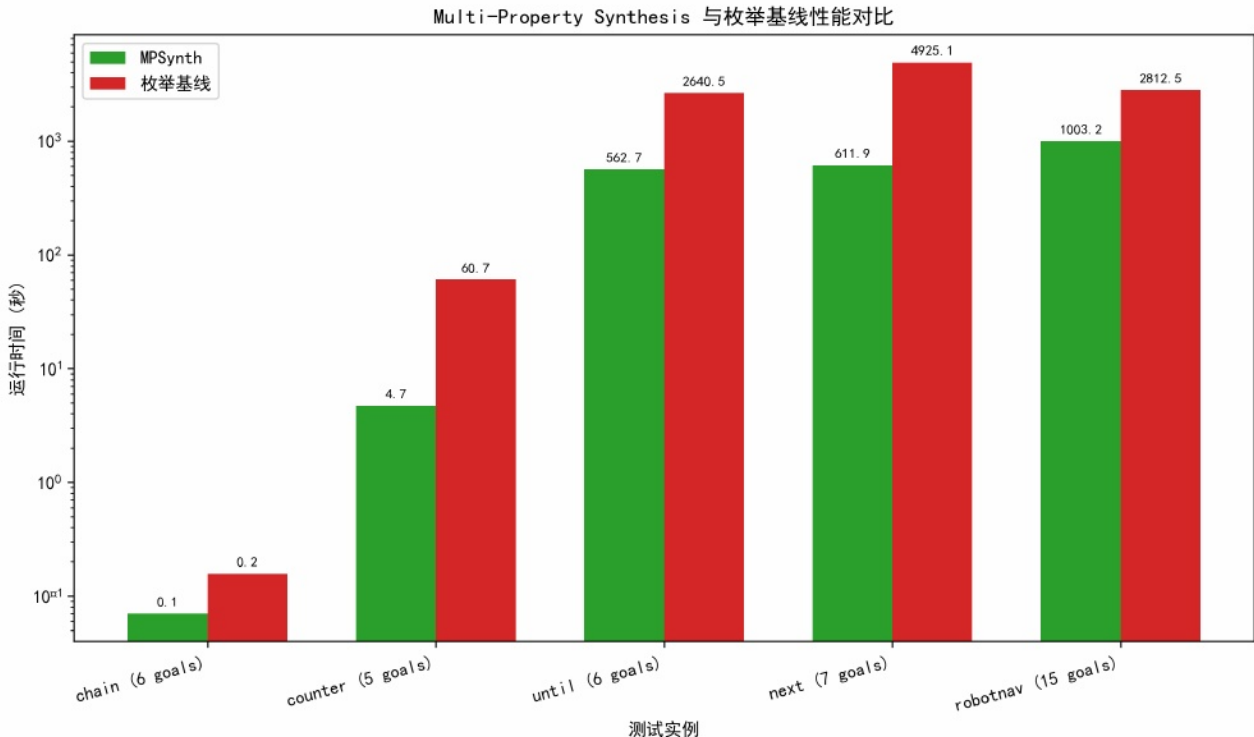
]

2.2 Symbolization advantages

By encoding with Boolean variables, the explicit enumeration of 2^n target subsets is avoided, and the monotonicity ensures the efficiency of the representation. Experiments show that this method can accelerate the process by up to two orders of magnitude compared to the enumeration baseline.

2.3 Performance comparison

The following figure shows the comparison of the running time between Multi-Property Synthesis (MPSynth) and the enumeration baseline on multiple benchmark instances (approximated based on the data in Table 1 of the paper, using a logarithmic scale).



3. Proc3D: Procedural 3D Generation and Parametric Editing

Proc3D addresses the issue of uneditable outputs from existing 3D generation models by proposing a representation method based on Procedural Compact Graph (PCG), enabling end-to-end generation from text to editable 3D models.

3.1 Procedural Compact Graph (PCG)

PCG is a lightweight graph representation where each node corresponds to a geometric operation (such as cube, cylinder, transformation, or merge), and the parameters are exposed as adjustable variables. Compared with Blender's native nodes or Python code, the token length of PCG is reduced by 4 to 10 times, and the compilation success rate is as high as 89%.

3.2 Generation and Editing Process

Generation: The user inputs a text prompt, and the LLM (such as GPT-4 or a fine-tuned LLaMA-3) generates the corresponding PCG, which is then converted by an interpreter into code executable by software like Blender.

- **Editors** can manually adjust parameters via a slider or instruct the LLM to modify the PCG through natural language commands, with real-time model updates. The editing speed can reach 0.01 seconds, which is 400 times faster than traditional methods.

3.3 Experimental performance

Proc3D achieved a ULIP score of 0.15, significantly outperforming baselines such as SDFusion (0.11) and AutoSDF (0.07), and is capable of generating complex objects not present in the training set, such as tigers and airplanes.

4. SpaceHMchat: Human-Machine Collaboration for Spacecraft Power System Health Management

SpaceHMchat proposes a full-loop health management framework based on human-machine collaboration to address the challenges faced by spacecraft power systems (SPS) in the era of satellite mega-constellations.

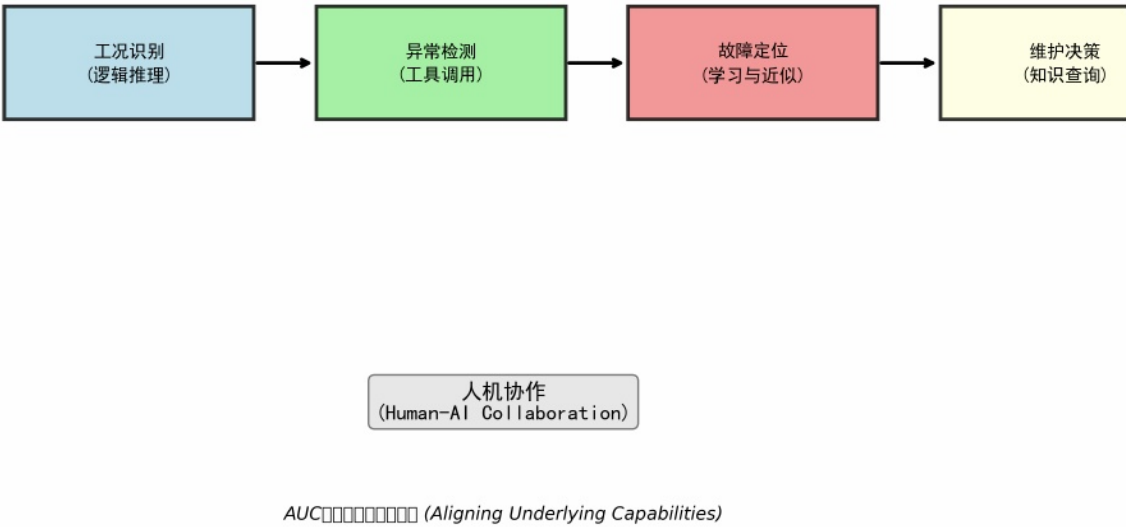
4.1 AUC Principle

(Aligning Underlying Capabilities) The principle of guidance aligns each sub-task of health management with the corresponding capabilities of the LLM:
- **Maintenance decision (knowledge-intensive)** → RAG + LLM reasoning
- **Work condition identification (logical reasoning)** → Prompt + CoT
- **Anomaly detection (tool-dependent)** → Function call + MCP
- **Fault localization (learning and approximation)** → fine-tuning + LoRA

4.2 System process

The following figure shows the four stages of SpaceHMchat and their interaction relationships.

SpaceHMchat 全回路健康管理流程示意图



4.3 Experimental verification

On the self-built XJTU-SPS dataset, SpaceHMchat achieved:

- Operating condition identification: 100% conclusion accuracy rate
- Anomaly detection: Tool invocation success rate > 99%
- Fault location: Accuracy rate > 99%
- Maintenance decision: Knowledge retrieval time < 3 minutes, citation accuracy rate 97.3%

5. Comprehensive comparison

Method	Core contributions	Key technologies	Limitations
Multi-Property Synthesis	Multi-objective synthesis	Symbolic points	The state space may still grow exponentially.
Proc3D	Editable 3D generation	representation, LLM	Generating and editing is limited to cubic primitives.
SpaceHMchat	Human-machine collaborative health management	AUC	Double-path reasoning increases overhead.

6. Future direction principle, RAG, fine-tuning

- The **deep integration of formalization and generation**: Directly incorporating **LTLf** constraints into the 3D generation process, to ensure that the generated objects satisfy temporal logic.
- **Automated knowledge base construction**: Utilizing **LLM** to automatically extract knowledge from design documents and fault reports, reducing the cost of manual maintenance.
- **Online adaptive planning**: Combining the **TIDE-class** planning method with SpaceHMchat to achieve dynamic task replanning.

References

- [1] Weinhuber, C., et al. (2026). Multi-Property Synthesis. *arXiv:2601.10651*.
- [2] Raji, F., et al. (2026). Proc3D: Procedural 3D Generation and Parametric Editing of 3D Shapes with Large Language Models. *arXiv:2601.12234*.
- [3] Di, Y., et al. (2026). Empowering All-in-Loop Health Management of Spacecraft Power System in the Mega-Constellation Era via Human-AI Collaboration. *arXiv:2601.12667*.